Experiment No: 2

Exploratory Data Analysis (EDA) – Data Import and Export

Aim

To import the KaggleV2-May-2016.csv dataset using Pandas, inspect and analyze the dataset, perform basic exploratory data analysis (EDA), and export the processed data into different file formats.

Algorithm

1. Import the required Python libraries.
2. Load the healthcare dataset (`KaggleV2-May-2016.csv`) using `pd.read_csv()`.
3. Display the first few records using `head()`.
4. Display the dataset information using `info()`.
5. Display statistical summary using `describe()`.
6. Display the column names.
7. Check the shape of the dataset.
8. Check for missing values.
9. Check for duplicate records.
10. Display unique values of each column.
11. Visualize the relationship between selected variables using a scatter plot.
12. Separate the feature variables (`X`) and target variable (`y`).
13. Export the processed dataset into CSV, Excel, and JSON formats.

```python
# Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
# Step 2: Upload Dataset in Google Colab
from google.colab import files
uploaded = files.upload()
```

Choose files KaggleV2-May-2016.csv

KaggleV2-May-2016.csv(text/csv) - 10739535 bytes, last modified: 23/07/2026 - 100% done

```
Saving KaggleV2-May-2016.csv to KaggleV2-May-2016 (1).csv
```

```python
print(df.head())
```

```
      PatientId  AppointmentID Gender          ScheduledDay  \
0  2.987250e+13        5642903      F  2016-04-29T18:38:08Z
1  5.589978e+14        5642503      M  2016-04-29T16:08:27Z
2  4.262962e+12        5642549      F  2016-04-29T16:19:04Z
3  8.679512e+11        5642828      F  2016-04-29T17:29:31Z
4  8.841186e+12        5642494      F  2016-04-29T16:07:23Z

         AppointmentDay  Age      Neighbourhood  Scholarship  Hipertension  \
0  2016-04-29T00:00:00Z   62    JARDIM DA PENHA            0             1
1  2016-04-29T00:00:00Z   56    JARDIM DA PENHA            0             0
2  2016-04-29T00:00:00Z   62      MATA DA PRAIA            0             0
3  2016-04-29T00:00:00Z    8  PONTAL DE CAMBURI            0             0
4  2016-04-29T00:00:00Z   56    JARDIM DA PENHA            0             1

   Diabetes  Alcoholism  Handcap  SMS_received No-show
0         0           0      0.0           0.0      No
1         0           0      0.0           0.0      No
2         0           0      0.0           0.0      No
3         0           0      0.0           0.0      No
4         1           0      0.0           0.0      No
```

```python
import os

print(os.listdir())
```

```
['.config', 'KaggleV2-May-2016.csv', 'sample_data']
```

```python
print("\nDataset Information")
df.info()
```

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 132084 entries, 0 to 132083
Data columns (total 14 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   PatientId       132084 non-null  float64
 1   AppointmentID   132084 non-null  int64
 2   Gender          132084 non-null  object
 3   ScheduledDay    132084 non-null  object
 4   AppointmentDay  132084 non-null  object
 5   Age             132084 non-null  int64
 6   Neighbourhood   132084 non-null  object
 7   Scholarship     132084 non-null  int64
 8   Hipertension    132084 non-null  int64
 9   Diabetes        132084 non-null  int64
 10  Alcoholism      132084 non-null  object
 11  Handcap         132083 non-null  float64
 12  SMS_received    132083 non-null  float64
 13  No-show         132083 non-null  object
dtypes: float64(3), int64(5), object(6)
memory usage: 14.1+ MB
```

```python
print("\nStatistical Summary")
print(df.describe())
```

```
Statistical Summary
          PatientId  AppointmentID            Age    Scholarship  \
count  1.320840e+05   1.320840e+05  132084.000000  132084.000000
mean   1.472340e+14   5.672981e+06      36.963569       0.098263
std    2.558273e+14   7.003391e+04      23.029998       0.297671
min    3.921784e+04   5.030230e+06      -1.000000       0.000000
25%    4.162841e+12   5.639690e+06      18.000000       0.000000
50%    3.155544e+13   5.678315e+06      37.000000       0.000000
75%    9.434637e+13   5.720900e+06      55.000000       0.000000
max    9.999816e+14   5.790484e+06     115.000000       1.000000

        Hipertension       Diabetes        Handcap   SMS_received
count  132084.000000  132084.000000  132083.000000  132083.000000
mean        0.196405       0.071008       0.022281       0.317936
std         0.397280       0.256839       0.161604       0.465676
min         0.000000       0.000000       0.000000       0.000000
25%         0.000000       0.000000       0.000000       0.000000
50%         0.000000       0.000000       0.000000       0.000000
75%         0.000000       0.000000       0.000000       1.000000
max         1.000000       1.000000       4.000000       1.000000
```

```python
print("\nColumns in Dataset")
print(df.columns.tolist())
```

```
Columns in Dataset
['PatientId', 'AppointmentID', 'Gender', 'ScheduledDay', 'Appointment
```

```python
print("\nShape of Dataset")
print(df.shape)

print("\nRows :", df.shape[0])
print("Columns :", df.shape[1])
```

```
Shape of Dataset
(132084, 14)

Rows : 132084
Columns : 14
```

```python
print("\nRandom Sample")
print(df.sample(5, random_state=42))
```

```
Random Sample
          PatientId  AppointmentID Gender          ScheduledDay  \
27185  4.821340e+12        5312204      F  2016-02-01T14:11:49Z
9679   7.652494e+14        5663412      F  2016-05-05T09:59:31Z
85880  9.341366e+13        5679791      F  2016-05-10T10:30:07Z
42838  3.256279e+13        5618944      M  2016-04-26T07:33:24Z
48930  5.113857e+13        5680977      F  2016-05-10T13:21:40Z

             AppointmentDay  Age   Neighbourhood  Scholarship  Hipertension  \
27185  2016-05-05T00:00:00Z   41     RESISTÊNCIA            0             0
9679   2016-05-06T00:00:00Z    2  FORTE SÃO JOÃO            0             0
85880  2016-05-13T00:00:00Z   55  JARDIM CAMBURI            0             1
42838  2016-05-05T00:00:00Z   40     SANTO ANDRÉ            0             0
48930  2016-05-10T00:00:00Z   36   SÃO CRISTÓVÃO            0             0

       Diabetes Alcoholism  Handcap  SMS_received No-show
27185         0          0      0.0           1.0      No
9679          0          0      0.0           0.0      No
85880         1          0      0.0           0.0      No
42838         0          0      0.0           1.0     Yes
48930         0          0      0.0           0.0      No
```

```
missing = pd.DataFrame({
    "Missing Values": df.isnull().sum(),
    "Percentage": (df.isnull().sum()/len(df))*100
})

print("\nMissing Values")
print(missing)
```

```
Missing Values
                Missing Values  Percentage
PatientId                    0    0.000000
AppointmentID                0    0.000000
Gender                       0    0.000000
ScheduledDay                 0    0.000000
AppointmentDay               0    0.000000
Age                          0    0.000000
Neighbourhood                0    0.000000
Scholarship                  0    0.000000
Hipertension                 0    0.000000
Diabetes                     0    0.000000
Alcoholism                   0    0.000000
Handcap                      1    0.000757
SMS_received                 1    0.000757
No-show                      1    0.000757
```

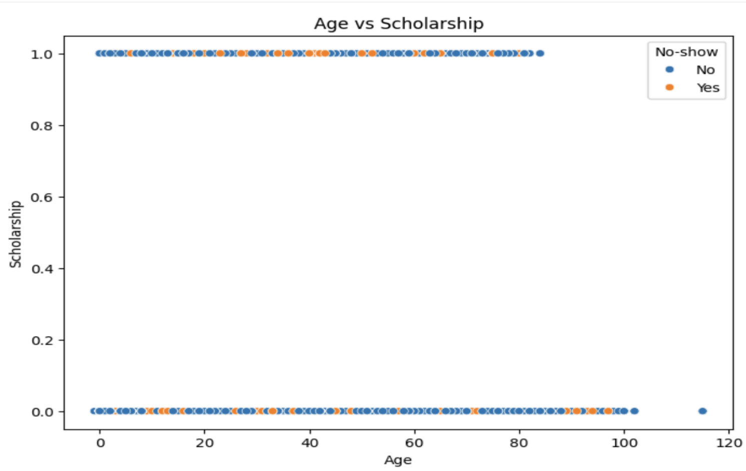
```
duplicates = df.duplicated().sum()

print("\nDuplicate Rows :", duplicates)
```

```
Duplicate Rows : 21556
```

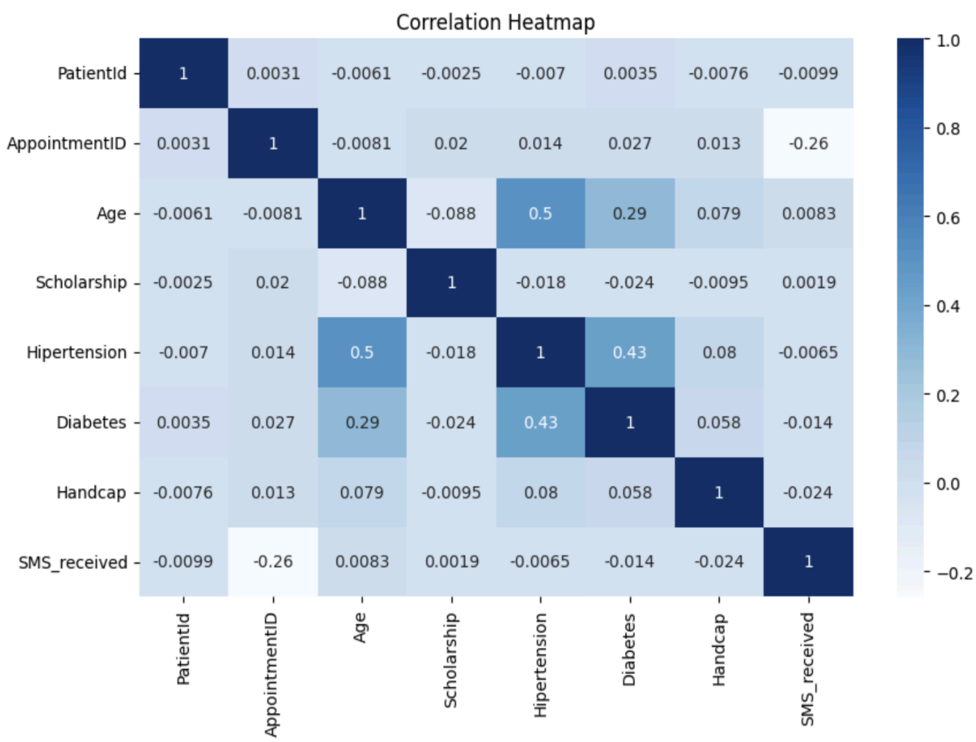
```python
print("\nUnique Values in Each Column\n")

for column in df.columns:
    print(column, ":", df[column].nunique())
```

```
Unique Values in Each Column

PatientId : 62299
AppointmentID : 110527
Gender : 2
ScheduledDay : 103550
AppointmentDay : 28
Age : 104
Neighbourhood : 82
Scholarship : 2
Hipertension : 2
Diabetes : 2
Alcoholism : 5
Handcap : 5
SMS_received : 2
No-show : 2
```

```python
plt.figure(figsize=(8,6))

sns.scatterplot(
    data=df,
    x="Age",
    y="Scholarship",
    hue="No-show"
)

plt.title("Age vs Scholarship")
plt.xlabel("Age")
plt.ylabel("Scholarship")

plt.show()
```



```
numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,6))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="Blues"
)

plt.title("Correlation Heatmap")

plt.show()
```



```
X = df.drop("No-show", axis=1)

y = df["No-show"]

print("\nFeature Data Types\n")
print(X.dtypes)

print("\nTarget Variable")
print(y.head())
```

```
Feature Data Types

PatientId           float64
AppointmentID         int64
Gender               object
ScheduledDay         object
AppointmentDay       object
Age                   int64
Neighbourhood        object
Scholarship           int64
Hipertension          int64
Diabetes              int64
Alcoholism           object
Handcap             float64
SMS_received        float64
dtype: object

Target Variable
0    No
1    No
2    No
3    No
4    No
Name: No-show, dtype: object
```

```
df.to_csv("Processed_KaggleV2.csv", index=False)

df.to_excel("Processed_KaggleV2.xlsx", index=False)

df.to_json("Processed_KaggleV2.json")

print("\nDataset Exported Successfully")
```

```
Dataset Exported Successfully
```

```
files.download("Processed_KaggleV2.csv")
files.download("Processed_KaggleV2.xlsx")
files.download("Processed_KaggleV2.json")
```

Downloading "Processed_KaggleV2.csv":
Downloading "Processed_KaggleV2.xlsx":
Downloading "Processed_KaggleV2.json":